

Group Index of 500 nm TE Silicon Strip Waveguides from Unbalanced MZIs, and an MZI-Coupled Ring Resonator

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Abstract

Design proposal for chip EBeam_rafael-w9v2. All circuits use the TE0 mode of a 500×220 nm Si strip waveguide. 5 Y-branch MZIs ($\Delta L = 25/50/75/100/200$ μm) give the group index (simulated $n_g = 4.218$ at 1550 nm). Two rings coupled through an MZI show wavelength-dependent, controllable coupling.

1 Introduction

n_g sets the FSR of rings and interferometers, the delay of delay lines and the length of phase shifters (e.g. thermo-optic). A coupling-controlled resonator sets the ring coupling with an interferometer phase instead of a fixed gap [1].

Objectives:

- $n_g(\lambda)$ of TE0 from the MZI FSR;
- $1/\text{FSR} \propto \Delta L$ over 5 values of ΔL ;
- MZI-coupled ring response vs. the MZI coupling.

2 Theory

2.1 Waveguide Compact Model

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (1)$$

2.2 MZI Transfer Function

Two ideal Y-branches [2]:

$$\frac{I_o}{I_i} = \frac{1}{4} \left| e^{-j\beta_1 L_1 - \frac{\alpha}{2} L_1} + e^{-j\beta_2 L_2 - \frac{\alpha}{2} L_2} \right|^2, \quad \beta = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}. \quad (2)$$

Identical, lossless arms, $\Delta L = L_2 - L_1$:

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta \Delta L)]. \quad (3)$$

2.3 Group Index from the FSR

Arm phase difference $\phi = 2\pi n_{\text{eff}} \Delta L / \lambda$. Adjacent minima differ by $\Delta\phi = 2\pi$, so $|d\phi/d\lambda| \text{ FSR} = 2\pi$ with

$$\frac{d\phi}{d\lambda} = -\frac{2\pi\Delta L}{\lambda^2} \left(n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \right) \Rightarrow \underset{\tilde{n}_g}{n_g} = \frac{\lambda^2}{\Delta L \cdot \text{FSR}}. \quad (4)$$

For several ΔL :

$$\frac{1}{\text{FSR}} = \frac{n_g}{\lambda^2} \Delta L, \quad \frac{\delta n_g}{n_g} \approx \sqrt{2} \frac{\delta \lambda_{\text{min}}}{\text{FSR}}. \quad (5)$$

2.4 MZI-Coupled Ring

A 2×2 MZI (two 3 dB couplers) replaces the ring coupler; one MZI output loops back to one input (round trip L_{rt}) [1], [3]:

$$t(\lambda) = |\sin(\beta \Delta L / 2)|, \quad T = \frac{a^2 - 2at\cos\theta + t^2}{1 - 2at\cos\theta + (at)^2}, \quad (6)$$

$a = e^{-\alpha L_{\text{rt}}/2}$, $\theta = \beta L_{\text{rt}}$. Critical coupling at $t = a$. A phase shift in one MZI arm tunes t .

3 Modelling and Simulation

3.1 Waveguide

Si strip, $w = 500$ nm, $h = 220$ nm, SiO_2 cladding, TE polarization (PDK type “Strip TE 1550 nm, w=500 nm” [4]). Lumerical MODE (FDE) [5]: Figs. 1, 2.

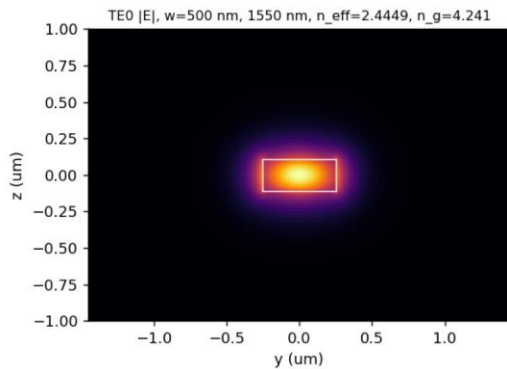


Fig. 1. TE0 mode profile ($|E|$) at 1550 nm, MODE (FDE).

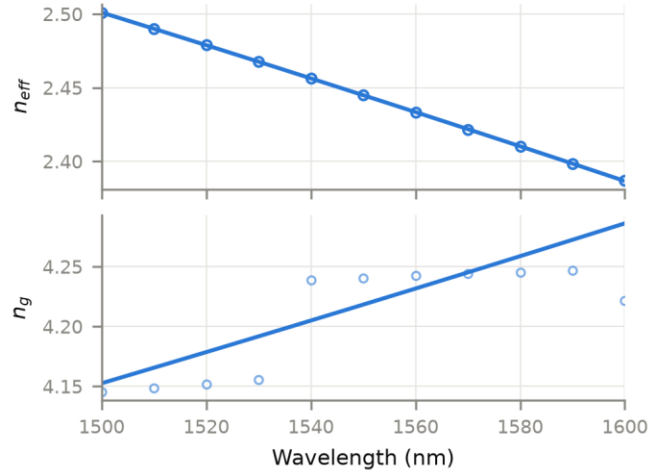


Fig. 2. n_{eff} and n_g vs. wavelength. Markers: MODE (FDE); lines: Eq. (7).

Compact model, λ in μm , $\lambda_0 = 1.55 \mu\text{m}$:

$$n_{\text{eff}}(\lambda) = 2.4449 - 1.1441(\lambda - \lambda_0) - 0.4310(\lambda - \lambda_0)^2. \quad (7)$$

$n_g = 4.152 / 4.218 / 4.286$ at 1500 / 1550 / 1600 nm.

3.2 MZI Design

TABLE 1. Expected FSR from Eq. (7)

Device	ΔL (μm)	FSR (nm) at $\lambda =$			pts/ FSR
		1500	1550	1600	
MZITEw500dL25	25	21.67	22.78	23.89	21674
MZITEw500dL50	50	10.84	11.39	11.95	10837
MZITEw500dL75	75	7.22	7.59	7.96	7225
MZITEw500dL100	100	5.42	5.70	5.97	5418
MZITEw500dL200	200	2.71	2.85	2.99	2709

ebeam_y_1550 split and combine. FSR < 50 nm and > 2700 points per FSR at 1 pm (Table 1).

3.3 Circuit Simulation

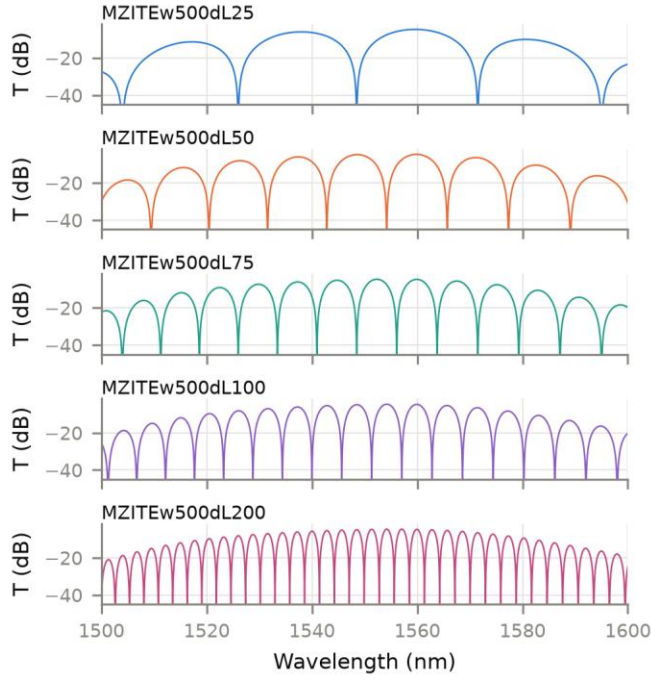


Fig. 3. Simulated MZI transmission, GC to GC: Eq. (3) with Eq. (7) and the PDK TE GC response [4].

Fig. 3.

3.4 Extraction Test on Simulated Spectra

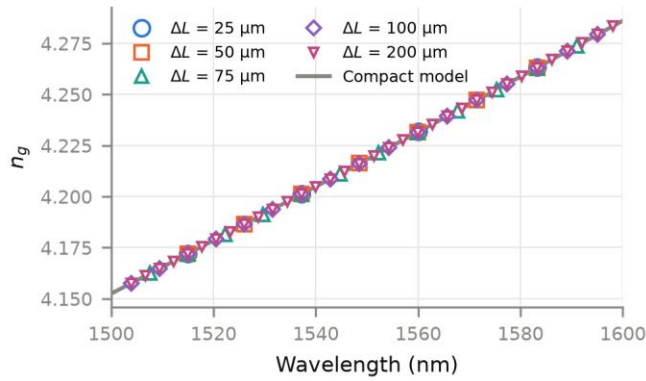


Fig. 4. n_g from the simulated FSR, Eq. (4) (markers), and Eq. (7) (line).

TABLE 2. n_g at 1550 nm extracted from Fig. 3

Device	FSR (nm)	n_g
MZITEw500dL25	22.781	4.2185
MZITEw500dL50	11.391	4.2183
MZITEw500dL75	7.594	4.2182

Device	FSR (nm)	n_g
MZITEw500dL100	5.696	4.2182
MZITEw500dL200	2.848	4.2182
Slope, Eq. (5), MZITEw500	4.2182	

Eqs. (4), (5) recover the model n_g (Fig. 4, Table 2).

3.5 Procedure for Measured Data

1. Subtract the LBTE spectrum (dB).
2. Locate the minima; quadratic fit around each null (sub-pm).
3. FSR of adjacent minima $\rightarrow n_g(\lambda)$, Eq. (4).
4. Slope over ΔL , Eq. (5); compare with Eq. (7).

3.6 MZI-Coupled Ring

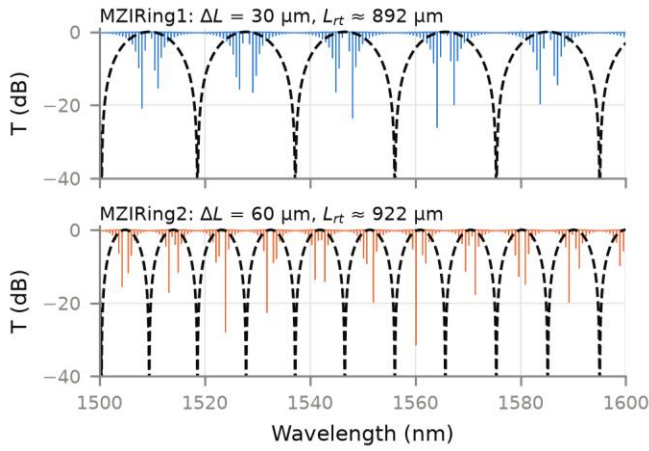


Fig. 5. MZI-coupled rings, Eq. (6), ideal couplers, 3 dB/cm assumed. Solid: ring through port; dashed: t^2 of the MZI.

TABLE 3. MZI-coupled ring parameters (1550 nm)

Device	ΔL (μm)	L_{rt} (μm)	FSR_{MZI} (nm)	FSR_{ring} (nm)	pts/ FSR
MZIRing1	30	892	19.0	0.639	639
MZIRing2	60	922	9.5	0.618	618

Two ebeam_bdc_te1550 couplers, loop $\approx 700 \mu\text{m}$. Resonance depth follows $t^2(\lambda)$, deepest near $t = a \approx 0.969$ (Fig. 5). Ring FSR $\approx 0.6 \text{ nm}$, > 600 points per FSR at 1 pm (Table 3). Measured: fit Eq. (6) per resonance $\rightarrow t, a$; compare $t(\lambda)$ with $|\sin(\beta\Delta L/2)|$ using n_{eff} from the MZIs.

3.7 Loopback

$IL_{GC} = 1/2 T_{LB}$ [dB]. Expected (PDK TE GC model [4]): LBTE peak -4.6 dB at 1555 nm, 3 dB bandwidth 47 nm.

4 Fabrication

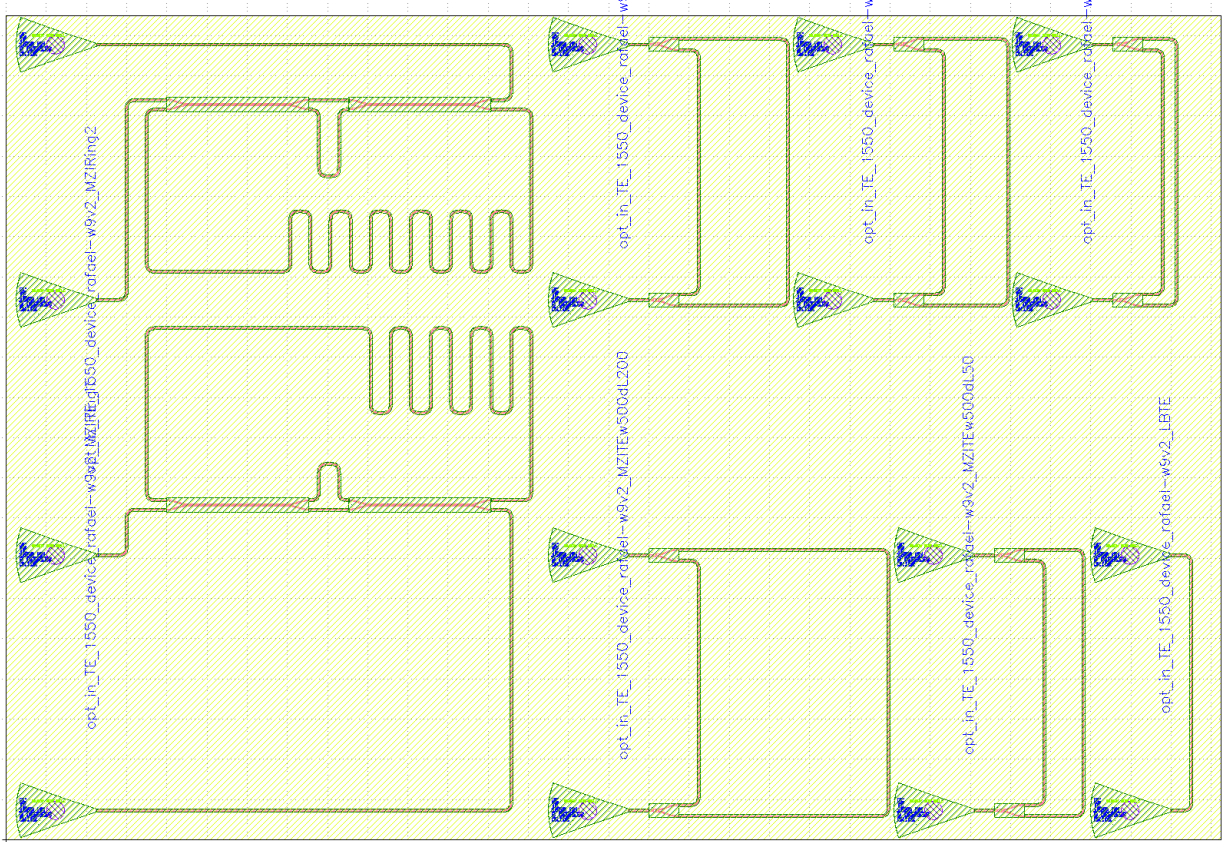


Fig. 6. Layout of *EBeam_rafael-w9v2* ($605 \times 410 \mu\text{m}$).

TABLE 4. Devices on *EBeam_rafael-w9v2*

Device	Pol.	Description
MZIRing1	TE	MZI-coupled ring, $2 \times bdc$, $\Delta L = 30 \mu\text{m}$, $L_{rt} \approx 892 \mu\text{m}$
MZIRing2	TE	MZI-coupled ring, $2 \times bdc$, $\Delta L = 60 \mu\text{m}$, $L_{rt} \approx 922 \mu\text{m}$
MZITEw500dL200	TE	MZI, Y-branch, $\Delta L = 200 \mu\text{m}$, $w = 500 \text{ nm}$
MZITEw500dL50	TE	MZI, Y-branch, $\Delta L = 50 \mu\text{m}$, $w = 500 \text{ nm}$
MZITEw500dL100	TE	MZI, Y-branch, $\Delta L = 100 \mu\text{m}$, $w = 500 \text{ nm}$
MZITEw500dL75	TE	MZI, Y-branch, $\Delta L = 75 \mu\text{m}$, $w = 500 \text{ nm}$
MZITEw500dL25	TE	MZI, Y-branch, $\Delta L = 25 \mu\text{m}$, $w = 500 \text{ nm}$
LBTE	TE	GC loopback, $L = 144 \mu\text{m}$

KLayout, SiEPIC-Tools [6], EBeam PDK [4] (Fig. 6, Table 4). TE GCs, 127 μm pitch; labels opt_in_TE_1550_device_rafael-w9v2_<name>. Functional check: 0 errors; DRC: 0 errors. Electron-beam lithography, Applied Nanotools NanoSOI [7], SiEPIC openEBL.

5 Experimental Data

To be completed (1500–1600 nm sweep, 1 pm step).

6 Analysis

To be completed.

7 Conclusion

To be completed.

References

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- [4] SiEPIC, “SiEPIC EBeam PDK, version 0.4.53.” https://github.com/SiEPIC/SiEPIC_EBeam_PDK, 2026.
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